My Report: The Results of the Timing Experiment

size of array (n)	Number of items sorted	Bubble Sort average time per array	Selection Sort average time per array
500	500,000	0.0322	0.0162
2500	2,500,000	0.9476	0.5138
5000	5,000,000	4.1898	1.7297

I have performed a timing experiment that compared the average time taken in seconds for each array to be sorted with the bubble sort algorithm and the selection sort algorithm. I was sorting through 1000 arrays for both sorts. The values of 'n' I used were 500, 2500, and 5000 in this increasing order. The number of items sorted with n = 500 was 500,000 per algorithm. The number of items sorted with n = 2500 was 2.5 million. And finally, the number of items sorted with n = 5000 was 5 million. I found after completing the line graph that the selection sort algorithm is faster than the bubble sort algorithm by over double the speed. For instance, with 5000 items to sort per array with 1000 arrays, the average time for the bubble sort was approximately 4.2 seconds while the average was 1.7 seconds for the selection sort. The graphs of the bubble and selection sort algorithms both have a moderate upward curve, resembling the graph of an O(n^2) algorithm which, with more data, would have a more defined shape. These two sorts have greater complexity than O(1) for example and their complexity growth is poor compared with O(1) (often called "constant time") and O(log(n)) which are both much faster.

Both algorithms have a complexity of O(n^2). When we analyze algorithms, we look at the rate of change over time to determine the efficiency of the algorithm to compute the outputs. Exponential Big-O talks of a rate of change by increasing factors over time where the performance of the algorithm is proportional to the square of the size of the input elements.

When we look at Big-O notation, it is like calculus when we are finding limits where we only care about the dominant term. This is usually the term with the highest degree since if n becomes sufficiently large, the other factors become insignificant. Since both the selection and bubble sort algorithms are of the time complexity O(n^2), the reason the selection sort is faster is because the worst-case time for the selection sort is generally quicker than the bubble sort. The nature of the selection sort algorithm involves leaving well-placed items where they already are as it sorts the list and while the bubble sort focuses solely on "bubbling up" the largest item to the end of the sequence of items, making the algorithm slower. This is consistent with the results I obtained in my timing experiment. The machine I used to run the experiments on is my personal laptop. It is a Dell Latitude E5470 running on Windows 10. Its processor is an Intel(R) Core(TM) i5-6300U CPU. It's a 64-bit operating system and it has 8.00 GB of RAM installed.